Finite daughter universes inside the Cauchy development of regular black holes

Project Baby Universe — Genesis Branch
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Abstract

In pure-gravity regularisations of black holes by quasi-topological curvature towers, a collapsing dust ball bounces in the regular core and re-expands. Its continuation through the inner horizon is not predicted by the field equations once perturbations turn the Cauchy horizon into a weak null singularity. We show that this continuation is not needed for a finite daughter. The bounced interior — a closed, homogeneous, expanding patch of universe hidden behind the event horizon — lies in the maximal globally hyperbolic development until the Cauchy horizon, which inside the ball is the ingoing null cone from the point where the ball's surface leaves the core. In the de Sitter-core limit this cone reaches the centre exactly when the core's de Sitter phase would end: the Cauchy horizon is the core's de Sitter horizon, and the conformal time available to the daughter is $\pi/2$ after the bounce (numerically $[\pi/2, 2.0]$). Large holes therefore host a daughter that inflates for $N \simeq \ln C/(D-1)$ e-folds inside the determined region. Near-threshold, compact collapse in $D \geq 7$ goes further: the daughter reaches its maximum and recollapses before its predictable history ends. With the Dymnikova tower in $D = 10$ it completes a full cycle, reaching its second bounce. These regimes are open sets of initial data.

1 Set-up

Model U is $D = 4 + n$ dimensional gravity with the quasi-topological tower $h(\psi)$ of Bueno, Cano and Hennigar [arXiv:2403.04827]: $\mu = r^{D-1}h(\psi)$, with Hayward $h = \psi/(1-\psi)$ or Dymnikova $h = -\ln(1-\psi)$, in units $l = \sqrt{\alpha} = 1$. A homogeneous dust ball of comoving radius χ_s has interior $ds^2 = -d\tau^2 + a^2(d\chi^2 + \sin^2\chi\, d\Omega^2_{D-2})$ with

$$\dot{a}^2 = a^2\psi(a) - 1, \qquad \psi = h^{-1}(C/a^{D-1}), \qquad C = \mu/\sin^{D-1}\chi_s, \tag{1}$$

and every closed universe bounces at $a_b \geq l$ (Oppenheimer–Snyder in these theories: [arXiv:2505.09680]). The exterior is the static hole $f = 1 - r^2\psi(\mu/r^{D-1})$ with horizons $r_- < r_+$ for $\mu > \mu_{\rm ext}$. The initial data (the ball at rest at $R_0 = a_{\max}\sin\chi_s > r_+$) are untrapped.

2 The daughter region

Lemma 1. *f decreases strictly on $(0, r_-]$. For Hayward, $f' = 2\mu r\big(\frac{D-3}{2}r^{D-1} - \mu\big)/(r^{D-1}+\mu)^2$ and r_- lies below the minimum of f; for Dymnikova this is checked numerically.*

Hence the surface ($\dot{r}^2 = E^2 - f$) has no turning point between the bounce and r_-. It leaves the core at an event E, which lies on the exterior Cauchy horizon $\mathcal{N} = \{v = \infty\}$.

Theorem 2. *Let $D = \{$post-bounce events of the interior with $\chi < \chi_s - (\eta - \eta_E)\}$, with η the conformal time. Then D lies in the maximal globally hyperbolic development of the initial data, and outside $J^-(\mathcal{I}^+)$ of the parent.*

Sketch. Inside the dust the future Cauchy horizon is generated by the ingoing null rays from E. Points beyond them have past causal curves leaving through the surface after E and running into the region beyond $\mathcal{N}$, which does not meet the initial slice. Post-bounce events lie inside the event horizon. □

Proposition 3 (de Sitter timing law)**.** *If $\psi \to 1/\alpha$ near the bounce and $r_- \to \sqrt{\alpha}$, then $\cos(\eta_E - \eta_b) = \sin\chi_s$ and the cone reaches the centre at $\eta_{\rm cone} - \eta_b = \pi/2$, the end of the core's de Sitter phase. Numerically $\eta_{\rm cone} - \eta_b \in [\pi/2, 2.00]$ over $D = 5$–10, both towers and $\mu/\mu_{\rm ext} = 1.05$–100; at $\mu = 10^4\mu_{\rm ext}$ it equals $\pi/2$ to 10^{-7}.*

3 Results

- *Inflating daughters* ($D = 5$, $\chi_s = 0.5$). The centre expands by $\times 9.0$, 16.0 and 50.9 for $\mu/\mu_{\rm ext} = 10$, 10^2 and 10^4, with $a_{\rm cone} \simeq 1.72\, C^{1/(D-1)}$ and $N_{\rm acc} \simeq \ln C/(D-1)$ accelerating e-folds inside D. Its predictable history ends while it is still expanding.

- *Recollapsing daughters.* The daughter's centre reaches its maximum inside D for 85% (Dymnikova, $D = 10$), 79% (Hayward, $D = 10$), 54% (Hayward, $D = 8$) and 0.4% (Hayward, $D = 5$) of initially untrapped configurations with $\mu/\mu_{\rm ext} \in [1.005, 30]$, $\chi_s \in [0.3, 1.565]$.

- *Full cycle.* Dymnikova, $D = 10$, $\mu \lesssim 1.1\mu_{\rm ext}$, $\chi_s \gtrsim 1.27$ (2.8% of the grid). Example: $\mu = 1.05\mu_{\rm ext}$, $\chi_s = 1.4$, starting at $R_0 = 1.024\, r_+$. The bounce is at $a_b = 1.009$; the maximum $a_{\rm max} = 1.119$ is reached at $\eta - \eta_b = 0.817$; the second bounce comes at 1.633; the cone reaches the centre at 1.690. In proper time the cycle lasts $1.73\, l$, about 10^{-28} s for collider parameters. Hayward towers do not reach the second bounce: their conformal half-period stays ≥ 1.19 ($D = 10$).

An ODE integration and an independent quadrature agree to 10^{-5}.

4 Robustness and limitations

- *Perturbations.* Mass inflation grows as $v \to \infty$; points of D receive influx only from finite v, so D minus a thin layer at its future boundary is stable at the level of the Ori model.

- *What happens beyond the cone is not claimed.* In $D = 5$ the Cauchy horizon is a Tipler-weak null singularity, so the matter reaches it with finite distortion but the continuation is not determined.

- *Realism.* The construction uses dust balls released from rest; the recollapse regimes require compact, near-threshold collapse. Collider holes form from partons (null shells), and pressure and semiclassical effects are not included.

- *Alternatives.*

  – A compact trapped region [Carballo-Rubio et al., arXiv:2607.03916] removes the Cauchy horizon, but the bounced matter then returns to the parent.

  – Evaporation drives $\kappa_- \to 0$ only as $v^{-1/2}$, which does not bound the blueshift.

  – Hyperbolic or toroidal horizons [arXiv:2505.04427] avoid Cauchy horizons but do not arise from spherical collapse.

Reproducibility

`CODE/babyuniverse/daughter12.py`, `CODE/scripts/round12.py`, `DATA/round12.json`, `CODE/tests/test_rou`

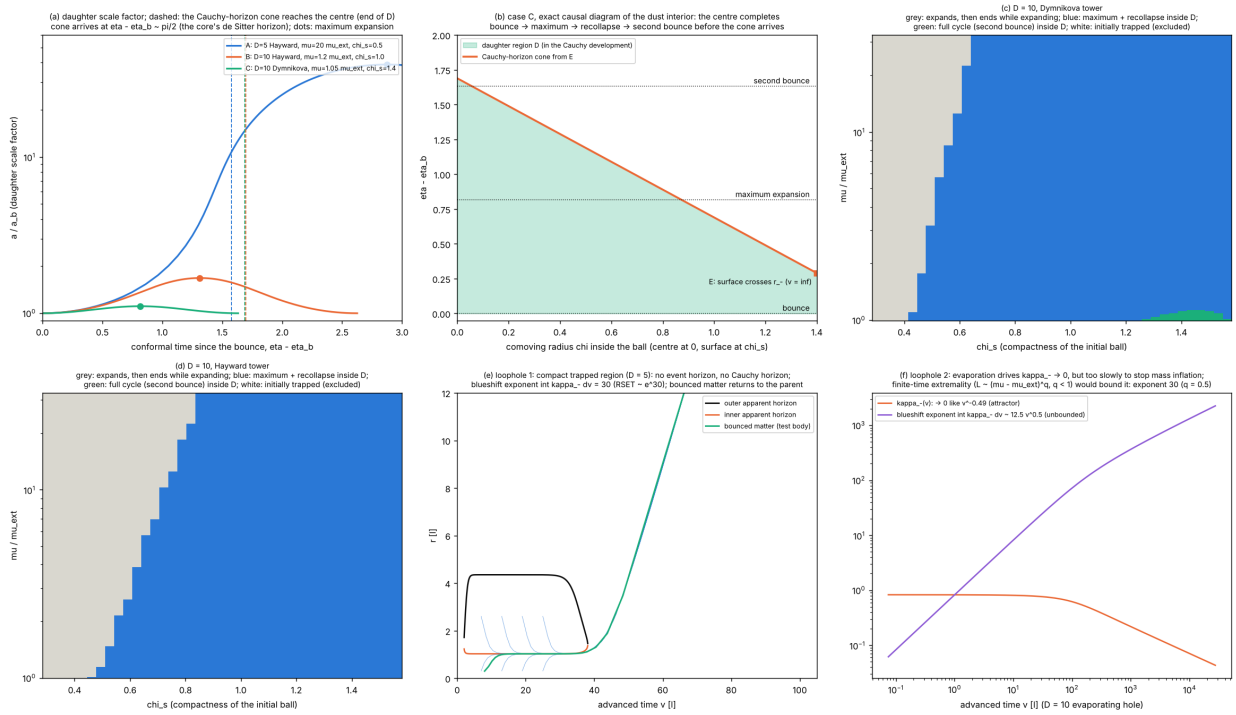


Figure 1: (a) Scale factor and cone arrival for three cases; (b) exact causal diagram of the interior for the full-cycle case; (c,d) regime maps for $D = 10$; (e,f) the compact-trapped-region and inner-extremality loopholes.

References

Bueno, Cano, Hennigar, PLB 861 (2025), arXiv:2403.04827. Bueno, Cano, Hennigar, Murcia, arXiv:2412.02740, 2412.02742. Bueno, Cano, Hennigar, Murcia, Vicente-Cano, arXiv:2505.09680. Carballo-Rubio, Di Filippo, Liberati, Visser, arXiv:2607.03916. Di Filippo, PRL 136, 161402 (2026), arXiv:2510.20649. Calzà, Rinaldi, Zerbini, PRD 112, 024024 (2025), arXiv:2505.04427. Gajic, Luk, Pure Appl. Anal. 1, 263 (2019), arXiv:1709.09137. McMaken, PRD 107, 125023 (2023), arXiv:2303.03562. Poisson, Israel, PRD 41, 1796 (1990). Ori, PRL 67, 789 (1991).

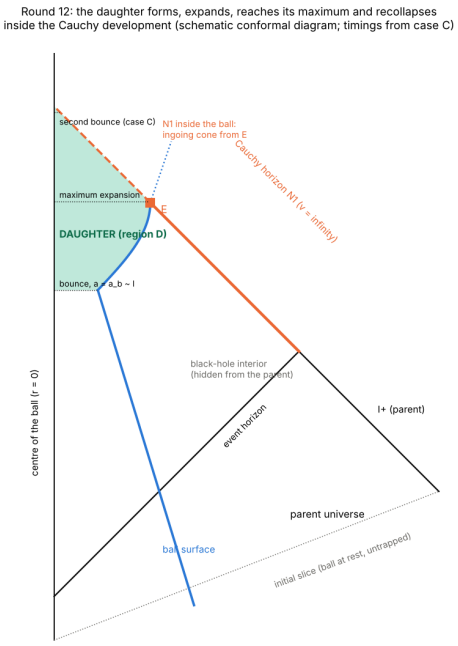


Figure 2: Schematic conformal diagram: the daughter region D lies below the Cauchy horizon $\mathcal{N}$ and its continuation as the cone from E.